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Cambodia: Utility-Scale Battery Energy Storage Project

Project Name
Utility-Scale Battery Energy Storage Project
Project Number
59110-001
Country / Economy
- Cambodia

Project Status
Proposed
Project Type / Modality of Assistance
- Grant
- Loan

Source of Funding / Amount

Grant: Utility-Scale Battery Energy Storage Project

Source	Amount
Asian Development Fund	US$ 5.00 million

Loan: Utility-Scale Battery Energy Storage Project

Source	Amount
Concessional ordinary capital resources lending	US$ 40.00 million
United Kingdom-ASEAN Catalytic Green Finance Facility Trust Fund	US$ 6.00 million
Green Climate Fund	US$ 12.44 million

Sector / Subsector
- Energy / Electricity transmission and distribution

Gender
Some gender elements
Description
The proposed project will finance the construction of a 250 megawatt (MW)/500 megawatt-hour (MWh) battery energy storage system (BESS) at the Takeo substation, south of Phnom Penh. It will strengthen power system resilience, improve electricity reliability, and enable higher integration of renewable energy into Cambodia's transmission network to improve system flexibility. These measures will support sustainable and inclusive economic growth and help the country meet its emission reduction targets. The BESS supports regional cooperation and integration by strengthening power trade and serving as a critical interface for electricity imports from southern Viet Nam. It improves cross-border system operations and improves grid stability, contributing to the ASEAN Power Grid (APG) and deepening transmission integration. The project supports digital transformation and private sector development by deploying a digitally controlled BESS that creates a more predictable and reliable environment for renewable energy investment in the country.
Project Rationale and Linkage to Country/Regional Strategy
In the decade preceding the coronavirus disease (COVID-19) pandemic, Cambodia was among the fastest-growing economies in Asia, with gross domestic product growth averaging 7.6% annually. Following a contraction in 2020, all major subsectors except real estate, construction, and tourism recovered, as the economy grew by 4.9% on average in 20212025. Growth is projected at 4.5% for 2026 with significant downside risk due to the ongoing conflict in the Middle East. Associated energy market disruptions pushed inflation to its highest level in over a decade in March 2026, with disproportionate impact on workers in the agriculture and transport sectors. Tourism is expected to face continued challenges because of geopolitical shocks and border restrictions.

In response to sustained economic growth and rising energy demand, Cambodia has made significant progress in expanding electricity generation capacity, access, and transmission infrastructure. Domestic power generation increased from 8.58 terawatt-hours (TWh) in 2020 to 20.56 TWh in 2025, driven largely by the addition of 1,668 MW of renewable energy capacity from 2020 to 2025. Electricity demand is projected to grow to 33.24 TWh by 2030 under a moderate growth scenario. Solar photovoltaic has emerged as the least-cost and fastest-growing technology, with installed capacity rising from 297 MW in 2020 to 1,476 MW in 2025. Reflecting this trajectory, the revised Power Development Master Plan (PDP), 20222040 targets developing up to 2,000 MW of utility-scale solar and 900 MW of wind capacity by 2030 to reduce dependence on fossil fuels and reinforce energy security for sustainable economic growth.

The rapid expansion of solar photovoltaic and wind capacity stresses the national power system. Grid congestion, daily and seasonal variability of renewable energy (including hydropower), and limited transmission grid flexibility are binding constraints. This is particularly evident in central and southern Cambodia where solar photovoltaic is concentrated. The Government of Cambodia has halted further financing of coal-fired power plants, while hydropower generation remains subject to significant seasonal variability. The rapid expansion of variable renewable energy and its inherent intermittency contribute to supplydemand imbalances and deterioration in power quality, including frequency deviations and voltage fluctuations. During periods of excess renewable power generation, this results in power curtailment. These constraints limit the capacity of the transmission system to integrate additional renewable energy capacity securely. They increase exposure to system disturbances and supply interruptions and undermine the reliability and quality of electricity supply. This leads to adverse effects on households and businesses, higher system costs, and constraints on inclusive economic growth and service delivery. The revised PDP, 20222040 envisaged 1,100 MW of BESS capacity by 2030. Electricit

du Cambodge (EDC)the state-owned utility responsible for generation, transmission, and distribution of electricity across Cambodiahas since identified the need to increase BESS capacity to more than 2,000 MW to support renewable energy integration and maintain transmission grid stability.

In response to the urgent need for renewable energy integration and grid stability, EDC is expanding BESS deployment. It manages the installation of 450 MW/900 MWh of BESS using its own funds. This is in addition to 100 MW/200 MWh under the ongoing Grid Reinforcement Project and the Energy Transition Sector Development Program (Subprogram 1) of the Asian Development Bank (ADB). Achieving the target of 2,000 MW of BESS capacity by 2030 requires EDC to strengthen institutional capability, regulatory frameworks, and operational readiness to fully manage a transmission system with BESS. Cambodia remains at an early stage in establishing technical standards, operational protocols, and regulatory arrangements for transmission-connected BESS. Institutional gaps include (i) absence of ancillary service regulations, (ii) limited domestic experience with grid-forming control and inverter-dominated system operation, and (iii) absence of a harmonized mechanism for operation and maintenance (O&M) of BESS. Strengthening institutional capacity within EDC is critical to ensure the safe, reliable, and financially sustainable deployment of BESS at scale. Women remain underrepresented in technical, engineering, and operational roles within EDC, particularly in transmission, generation, and substation-based operations. Women have limited access to advanced technical training and comprised only 5.6% of total participants in EDC's general technical training programs in 2025. This underscores the importance of targeted capacity-building initiatives to support inclusive participation in emerging energy storage functions.

Impact
Economic growth enabled through higher integration of renewable energy into the power system

Outcome
Transmission grid system security and capability enhanced

Outputs
Utility-scale BESS installed.

Technical and regulatory capacity of EDC enhanced.

Geographical Location
Nation-wide

Safeguard Categories

Environment
B
Involuntary Resettlement
C
Indigenous Peoples
C

Summary of Environmental and Social Aspects

Environmental Aspects
Involuntary Resettlement
Indigenous Peoples

Stakeholder Communication, Participation, and Consultation

During Project Design
During Project Implementation

Contact

Responsible ADB Officer
Bhattarai, Umang
Responsible ADB Department
Sectors Department 1
Responsible ADB Division
Energy Sector Office (SD1-ENE)
Executing Agencies
Electricite Du Cambodge

Timetable

Concept Clearance
01 Feb 2027
Fact Finding
19 Jan 2026 to 28 Jan 2026
MRM
-
Approval
-
Last Review Mission
-
Last PDS Update
23 Jul 2025

Funding

Project Page	https://www.adb.org//projects/59110-001/main
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